

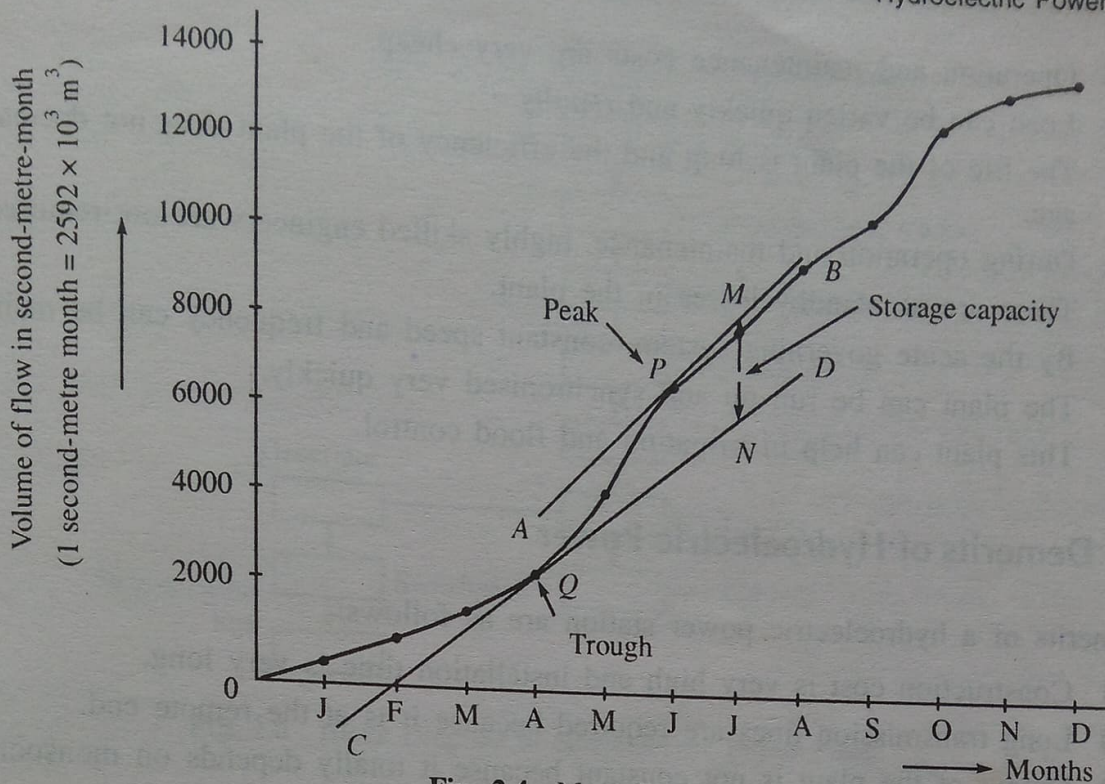


Fig. 2.4 Mass Curve.

Suppose, the mass curve is as shown in Fig. 2.5. Then, the trough points are found at B and C. RS will be the tangent at B and C. The peak point is A, i.e., at the top. Then, at A, a tangent AX will be drawn parallel to RS and the vertical distance between AX and SR will be the storage capacity.

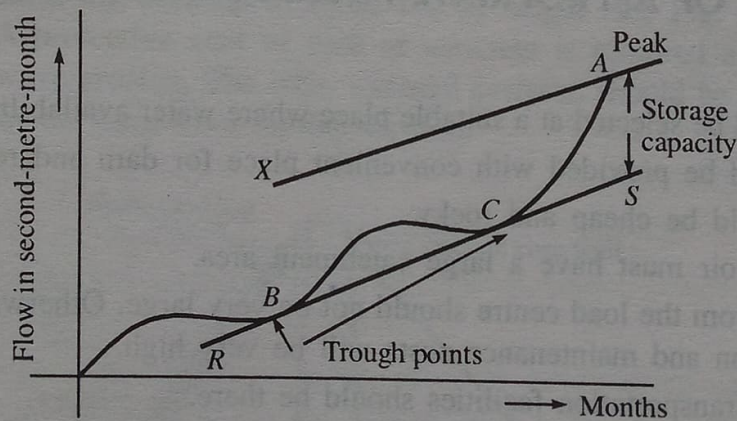


Fig. 2.5 Different Mass Curve.

2.3 MERITS AND DEMERITS OF HYDROPOWER STATION

2.3.1 Merits of Hydropower Station

The merits of a hydropower station are as follows:

- No fuel is required by a hydropower station.
- It is a very neat and clean plant.

- (c) Operation and maintenance costs are very cheap.
- (d) Load can be varied quickly and rapidly.
- (e) The life of the plant is long and the efficiency of the plant does not decrease with age.
- (f) During operation and maintenance, highly skilled engineers are not required.
- (g) There are no standby losses in the plant.
- (h) By the acute governing system, constant speed and frequency can be maintained.
- (i) The plant can be run up and synchronised very quickly.
- (j) This plant can help in irrigation and flood control.

2.3.2 Demerits of Hydroelectric Power

The demerits of a hydroelectric power station are as follows:

- (a) Construction cost is very high and installation time is very long.
- (b) Long transmission lines are required because it is at the remote end.
- (c) Output of the plant is not constant because it totally depends on monsoon.
- (d) It needs a large area.
- (e) Sometimes during the time of flood, excess water is to be drained out and that creates a problem for the nearby inhabitants.

2.4 SELECTION OF APPROPRIATE PLACE FOR HYDROELECTRIC POWER PLANT

- (a) Site should be selected at a suitable place where water availability will be very high.
- (b) Site should be provided with convenient place for dam and reservoir erection.
- (c) Land should be cheap and rocky.
- (d) The reservoir must have a large catchment area.
- (e) Distance from the load centre should not be very large. Otherwise, the transmission line erection and maintenance costs will be very high.
- (f) Adequate transportation facilities should be there.
- (g) There should be stream diversion route during the time of installation.

2.5 BLOCK DIAGRAM OF THE COMPONENTS OF HYDROELECTRIC POWER PLANT

Block diagram of the components of a hydroelectric power plant is shown in Fig. 2.6.

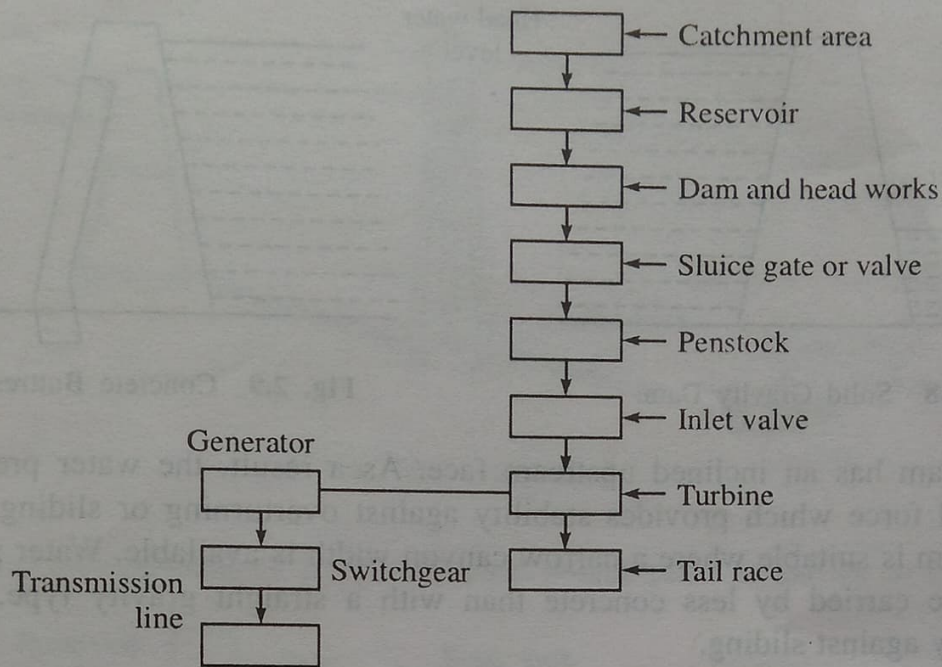


Fig. 2.6 Block Diagram of Hydroelectric Power Plant.

2.5.1 Some Important Elements of Hydroelectric Plant Dam

Hydroelectric plant dam is classified according to the structural materials. Its main purpose is to store water and to create head of water. Timber and steel are used for dams of height 6 m to 12 m. Earth dams are used up to about 100 m height (see Fig. 2.7). It is used for pervious foundation. A protective coat of rock or concrete is required at the water line to protect against water wave erosion. The other exposed surfaces should be covered with grass or vegetation to protect the dam from rainfall erosion.

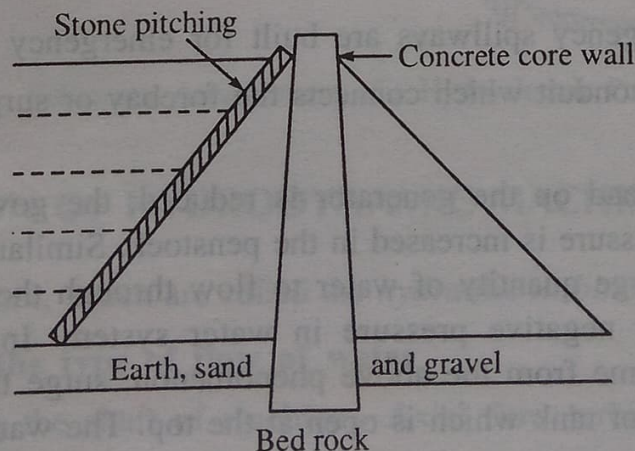


Fig. 2.7 Earth Dam.

In this dam, the seepage loss is more than other dams.

The masonry dams are of three major classes—*solid gravity dam* (see Fig. 2.8), *buttress dam* (see Fig. 2.9) and *arched dam*.

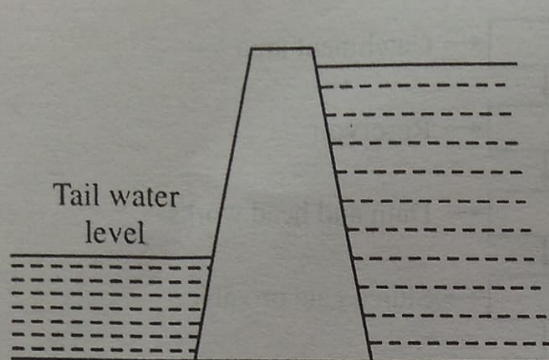


Fig. 2.8 Solid Gravity Dam.

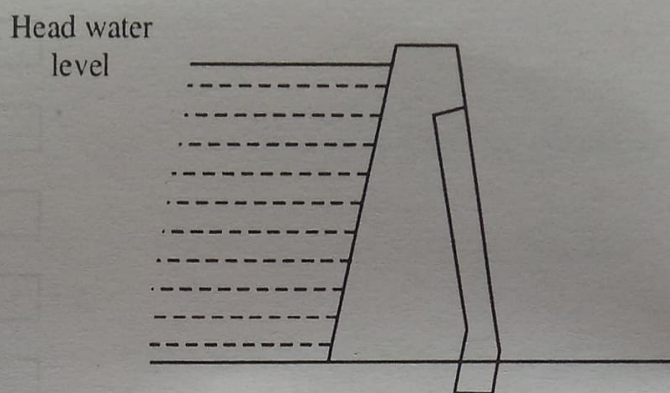


Fig. 2.9 Concrete Buttress Dam.

Buttress dam has an inclined upstream face. As a result, the water pressure creates a large downward force which provides stability against overturning or sliding.

Arched dam is suitable where a narrow canyon width is available. Water pressure against the arch will be carried by less concrete than with a straight gravity type. This dam has inherent stability against sliding.

Forebay. Forebay is considered as an enlarged body of water just above the intake for storing water temporarily to meet the hourly load fluctuations. This may be either a pond behind the diversion dam or an enlarged section of a canal.

Spillway. It acts as a safety valve. It discharges the overflow of water to the downstream side when the reservoir is full. This condition mainly arises during flood periods. Spillways can be classified as follows:

- Spillway dams or overflow spillways,
- Chute spillways,
- Side channel spillways,
- Shaft spillways, and
- Siphon spillways.

In some cases, emergency spillways are built for emergency purpose.

Penstock. It is a closed conduit which connects the forebay or surge tank to the scroll case of the turbine.

Surge tank. When the load on the generator is reduced, the governor tends to close the turbine gates and water pressure is increased in the penstock. Similarly, in case of larger load, the gate opens to allow large quantity of water to flow through the penstock. There will be a tendency of vacuum or negative pressure in water system. In case of closed conduit penstock, to protect the same from the above phenomenon, surge tank is provided. A surge tank is the small reservoir or tank which is open at the top. The water level in the surge tank rises or falls to control the pressure variation. During more pressure, water hammering occurs in the penstock, which damages it if surge tank is not provided. Figure 2.10 describes the arrangement of surge tank.

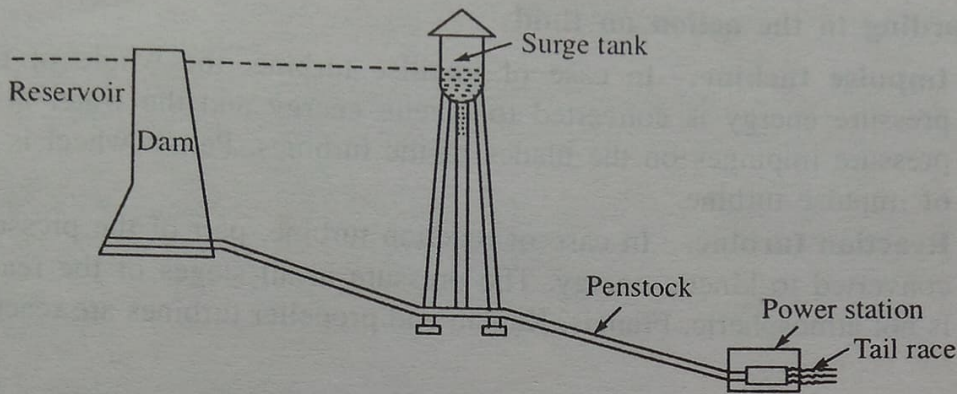


Fig. 2.10 Surge Tank.

A schematic arrangement of a hydroelectric power plant is shown in Fig. 2.11.

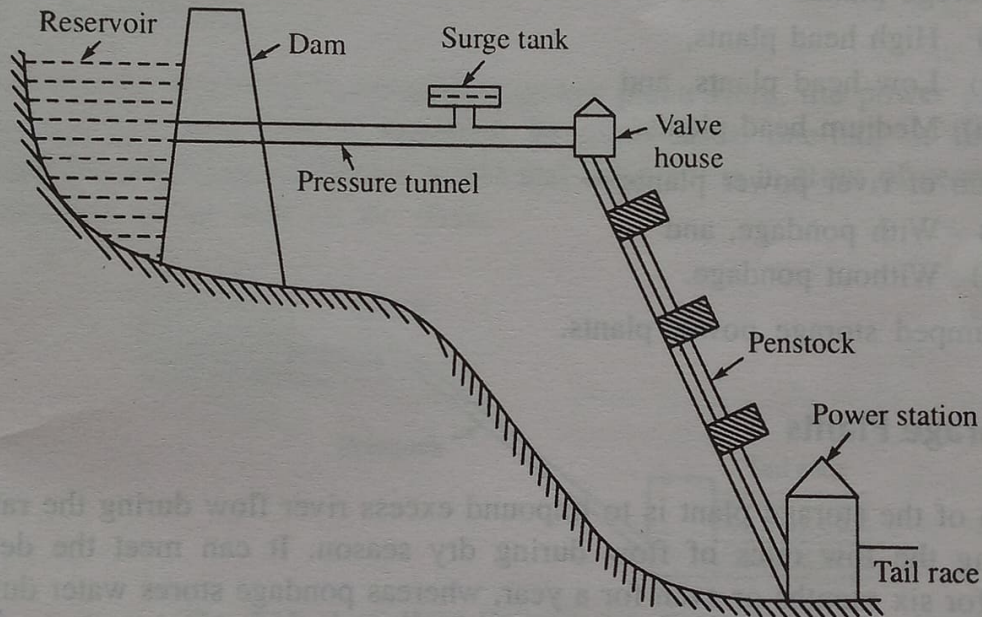


Fig. 2.11 Schematic Arrangement of a Hydroelectric Power Plant.

2.6 CLASSIFICATION OF HYDRODYNAMIC MACHINES

The hydrodynamic machines, which are called the *hydraulic turbines*, are classified as follows:

(a) According to the type of flow of water

- (i) Axis along the shaft of machine—Axial flow turbines (Propeller turbine and Kaplan turbine).
- (ii) Axis along the radius—Radial flow turbines (Francis turbine).
- (iii) Axis along the tangential directions—Tangential flow turbines (Pelton wheel).
- (iv) Mixed flow (Radial inlet and axial outlet)—Francis turbine.

Nuclear Power Plant

4.1 INTRODUCTION

The constituents of nuclear power plant are as follows:

- | | |
|----------------------|---------------------|
| (a) Nuclear reactor, | (b) Heat exchanger, |
| (c) Steam turbine, | (d) Condenser, and |
| (e) Alternator. | |

The schematic diagram of nuclear power plant is shown in Fig. 4.1.

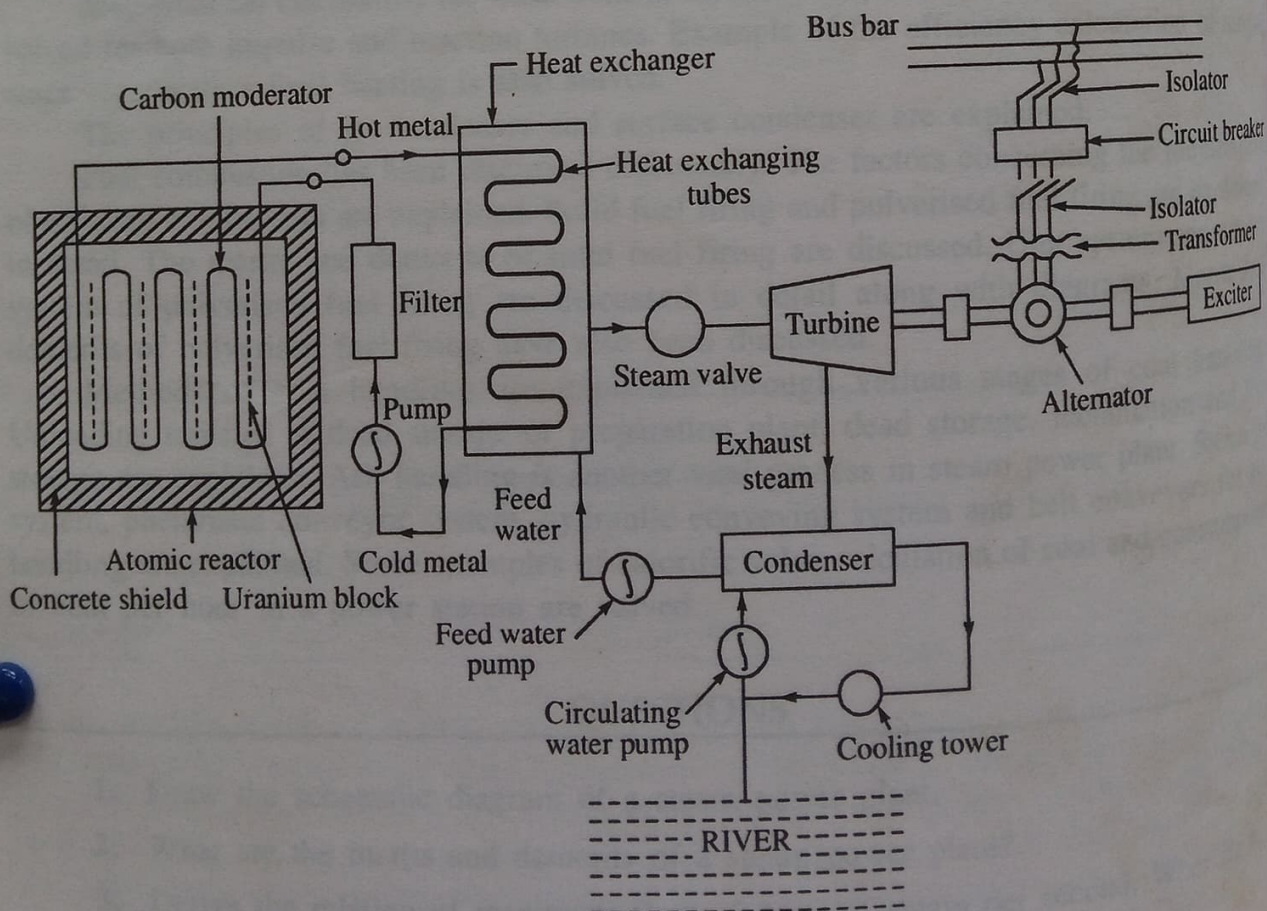


Fig. 4.1 Schematic Diagram of Nuclear Power Plant.

The purpose of a nuclear reactor is to generate heat. The nuclear heat is produced by splitting of nucleus. The process is termed as *nuclear fission*. For example, fusion of two heavy hydrogen atoms produces helium and neutron. The materials which undergo nuclear fission are uranium (U^{235}), thorium (Th^{232}) and plutonium (Pu^{239}). The subscript number is the atomic mass number. Light elements are fused by electrical discharge. The phenomenon is termed as *fusion*. Heavier atoms split to release energy. This phenomenon is termed as *fission*. The nuclear reactor consists of the following seven components (Fig. 4.2): (a) Fuel, (b) Moderator, (c) Reflector, (d) Control, (e) Coolant, (f) Shielding, and (g) Structure.

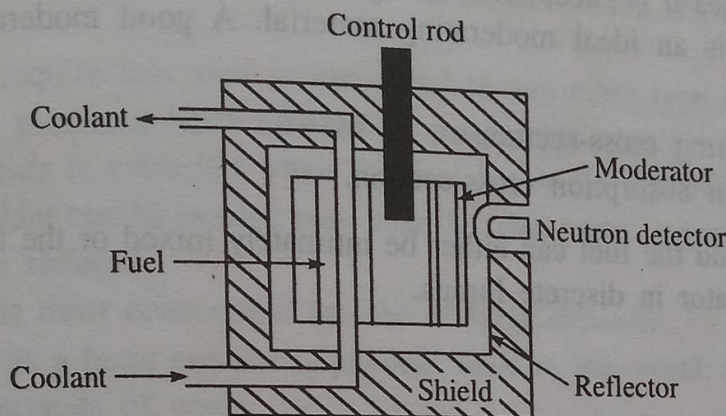


Fig. 4.2 Components of Nuclear Reactor.

4.1.1 Fuel

The reactor consists of U_{92}^{238} , Th_{90}^{232} , etc. as nuclear fuel. That means the atomic number (number of protons) of uranium is 92 and the atomic mass (number of protons + number of neutrons) is 238. Similarly, the atomic number of thorium is 90 and the atomic mass is 232. The present reactor uses uranium enriched with U_{92}^{235} along with aluminium or zirconium for dilution. In the nuclear reactor, tremendous amount of heat is generated in breaking of atoms of uranium or other similar metals of large atomic weight into metals of lower atomic weight by fission process. Then the heat is extracted by pumping fluid or metal-like liquid sodium or gas through the pipe. The fuel used in nuclear reactor is termed as natural uranium, that consists of 99.23% of U_{92}^{238} , 0.71% of U_{92}^{235} , and 0.06% of U_{92}^{239} . When the nucleus of U_{92}^{235} is bombarded by a neutron, two elements of lower mass number krypton and barium are formed along with a release of energy and detachment of two or three neutrons at a high speed of 2×10^7 m/s. For fission of U^{235} , 200 MeV energy is released ($1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$). The released two or three neutrons may cause fission reaction with other U^{235} nuclei. When such a reaction is controlled, a chain reaction occurs. But, if the chain reaction is not controlled, then an atom bomb may be produced. The splitting of 1 kg of uranium atoms yields 25×10^6 kWh in heat form.

The energy obtained by fission of 1 kg of uranium is approximately equivalent to burning of 2500 tonnes of coal. About 6.5×10^6 kWh electrical energy is developed from 1 kg of uranium.

4.1.2 Moderator

The objective of moderator is to reduce the neutron speed. When the fast neutrons collide with the nuclei of moderator material, they lose their energy and are slowed down. If a neutron collides with the nucleus of equal mass, it will lose all its energy, but if it collides with a much heavier nucleus, then it will lose very little energy. Hence the elements with smaller molecular weight are suitable for moderator material. For example, hydrogen, deuterium, helium, lithium, beryllium, boron, carbon, nitrogen and oxygen. Carbon is used in most of the large power producing reactors since it is cheap, satisfactory in operation and readily available in desired purity. Heavy water is an ideal moderating material. A good moderator should have the following properties:

- (a) High scattering cross-section.
- (b) Low neutron absorption cross-section.

The moderator and the fuel can either be intimately mixed or the fuel may be scattered throughout the moderator in discrete lumps.

4.1.3 Reflector

The reflector surrounds the reactor core within the thermal shielding arrangement. Most of the neutrons that want to escape from the fuel core are bounced back. Hence these neutrons reduce their speed and the low speed neutrons help in chain reaction. The reflector should also have good neutron scattering properties and small tendency to absorb neutrons. That is why, sometimes, the same material is used both for moderator and reflector.

4.1.4 Control Rods

The objective of the control rods is to regulate the fissioning in the reactor by absorbing the excess neutrons. The control rods are generally made of boron and are inserted into the reactor core from the top of the reactor vessel. If the fissioning rate is to be increased, the control rods are moved out slightly so that less number of neutrons will be absorbed by them. Similarly, to reduce fissioning rate, the control rods are to be moved slightly in.

4.1.5 Thermal Shielding

Thermal shielding surrounds the entire reactor core. It absorbs some of the radiations in the form of α , β particles, γ rays and escaping neutrons developed by fissioning. The shielding is usually constructed from iron. During above absorptions, it is heated. Coolant generally flows over the shielding to take away the heat.

4.1.6 Coolant

It is the medium through which the heat developed in the reactor is transferred to the heat exchanger for further use in power generation. The following materials are used as coolant:

- (a) Air, helium, hydrogen and CO_2 —gases.
- (b) Light and heavy water—liquids.
- (c) Molten sodium and lithium—metals.

4.2 MERITS AND DEMERITS OF NUCLEAR POWER PLANTS

The merits of nuclear power plants are as follows:

- (a) The amount of fuel required is quite small. Hence there is no problem of transportation.
- (b) The plants require less area as compared to any other type of the same size.
- (c) It not only produces large amount of power, but also produces valuable fissile material which is extracted when the fuel has to be renewed.
- (d) Since this plant can be located very near the load centre, primary power distribution cost can be reduced.
- (e) This plant is most economical in case of large capacity.
- (f) Since there is a large amount of nuclear fuels in the world, nuclear plants can be run for thousands of years.
- (g) The control of the output is extremely flexible.

The demerits of the nuclear power plants are as follows:

- (a) The initial capital cost is very high.
- (b) Technical knowledge is required for erection and commissioning.
- (c) There may occur a dangerous amount of radioactive pollution.
- (d) It is not well suited for varying loads.
- (e) The fuel is expensive and is difficult to recover.
- (f) Maintenance charge is high since large number of specially trained personnel are required.
- (g) The disposal of the radioactive products is a big problem.

4.3 SELECTION OF SITE OF NUCLEAR POWER PLANTS

The factors to be considered for selecting the nuclear power plant are as follows:

- (a) The site selected for the plant should be near a river, lake or seaside.
- (b) For the danger of presence of radioactivity in the atmosphere near the plant, there should be a reasonable distance between the nuclear power plant and the nearest populated area.
- (c) Transportation facilities are required during the construction stage.
- (d) The plants should be located as near to the load centre as possible, considering other factors.
- (e) The site should have adequate space for disposal of radioactive waste.
- (f) The land should have the supporting strength to carry reactors having weight of 1,00,000 tonnes and to impose bearing pressure 50 tonnes/m².